

QuantLab

Technical Documentation

Mathematical Foundations, Algorithms & Implementation Guide

For Quantitative Analysts, Portfolio Managers & Developers

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1. Architecture Overview

QuantLab is a comprehensive quantitative finance analytics platform built on a modern Python stack. It integrates real-time market data, advanced mathematical models, machine learning algorithms, and interactive visualizations into a single application designed for institutional investment research.

Purpose & Role in the Investment Workflow

The Architecture layer is the foundation that makes every other module possible. Its purpose is to provide a unified, high-performance data pipeline that fetches, caches, and distributes market data to all downstream analytics engines simultaneously. Without this shared data infrastructure, each module would independently fetch data, leading to inconsistent timestamps, redundant API calls, and misaligned analysis windows. The architecture ensures that when a user selects tickers and a date range in the sidebar, every module — from valuation to sentiment — operates on an identical, synchronized dataset.

Cross-Module Interactions

- **Architecture to Market Data (Section 2):** The data fetching pipeline supplies OHLCV DataFrames and computed returns to the Market Data module, which then calculates performance metrics consumed by Valuation, Portfolio, and ML modules.
- **Architecture to All Modules:** The `@st.cache_data` caching layer ensures that switching between tabs (e.g., from Technicals to Options) does not re-download data. Session state preserves user selections across tab navigation.
- **Architecture to Export (Section 13):** The export system collects computed results from all modules via session state, meaning the architecture's state management directly determines what data is available for PDF/Slides/Excel generation.
- **Architecture to Configuration (Section 14):** Sidebar widget values flow into every module's parameters — Monte Carlo simulation counts, ML training windows, clustering methods, and export section toggles are all routed through the architecture's configuration pipeline.

1.1 Technology Stack

Component	Technology	Role
Frontend	Streamlit	Interactive web UI with reactive widgets
Data Layer	yfinance / pandas	Market data acquisition & time-series management
Computation	NumPy / SciPy	Linear algebra, optimization, statistical tests
Machine Learning	scikit-learn	Predictive models, clustering, anomaly detection
Visualization	Plotly / Matplotlib	Interactive charts & static export images
Export	fpdf2 / xlsxwriter	PDF reports, slide decks, Excel workbooks

Component	Technology	Role
Technical Analysis	ta library	SMA, EMA, MACD, RSI, Bollinger Bands

1.2 Data Flow Architecture

The platform follows a pipeline architecture with five stages:

- **Sidebar Configuration** — Users select tickers, date ranges, analysis parameters, and model settings via Streamlit sidebar widgets.
- **Data Fetching** — Market data is retrieved from Yahoo Finance using `yf.download()` with MultiIndex handling for multi-asset portfolios. Results are cached via `@st.cache_data` to minimize API calls.
- **Analysis Engine** — Raw price/volume data flows through parallel computation paths: valuation models, portfolio optimization, bubble detection, Monte Carlo simulation, technical analysis, ML models, and sentiment analysis.
- **Visualization** — Results are rendered as interactive Plotly charts, data tables, and KPI cards within expandable Streamlit containers.
- **Export** — Users can generate PDF reports, slide decks, or Excel workbooks containing all computed metrics and visualizations.

1.3 Supported Asset Classes

Asset Class	Suffix / Format	Examples
Stocks / ETFs	Standard ticker	AAPL, MSFT, SPY, QQQ
Forex	=X suffix	EURUSD=X, GBPUSD=X, JPYUSD=X
Commodities	=F suffix	GC=F (Gold), CL=F (Oil), SI=F (Silver)
Cryptocurrency	-USD suffix	BTC-USD, ETH-USD, SOL-USD
Options	Equity ticker chains	Any equity with listed options

1.4 Session State & Caching

QuantLab leverages Streamlit's session state for persistent cross-interaction storage and `@st.cache_data` decorators for expensive computations. The caching strategy ensures that repeated parameter configurations do not trigger redundant API calls or recomputation of portfolio optimizations.

Code: Main Application Entry Point

```
def main():
    st.set_page_config(page_title='QuantLab', layout='wide')
    init_session_state()
    render_sidebar() # ticker, dates, parameters
    data = fetch_market_data(tickers, start, end)
    metrics = compute_performance(data)
```

```
valuations = run_valuation_models(data, metrics)
portfolio = optimize_portfolio(data, metrics)
render_dashboard(metrics, valuations, portfolio)
render_export_panel()
```

2. Market Data & Performance Metrics

QuantLab retrieves historical OHLCV (Open, High, Low, Close, Volume) data from Yahoo Finance using the `yfinance` library. The data layer handles multi-ticker downloads, MultiIndex DataFrame management, and automatic frequency alignment.

Purpose & Role in the Investment Workflow

Market Data is the central nervous system of QuantLab. Every analytical module depends on the price data and performance metrics computed here. The Sharpe ratio, volatility, and drawdown calculations provide the raw material for portfolio optimization constraints, valuation discount rates, and ML feature engineering. For institutional investors, standardized performance metrics enable apples-to-apples comparison across asset classes — comparing a forex pair's risk-adjusted return against an equity ETF requires consistent annualization and risk-free rate benchmarking.

Cross-Module Interactions

- **Market Data to Valuation (Section 3):** Daily returns feed directly into beta calculation (CAPM), which cascades into WACC and DCF discount rates. The risk-free rate fetched here anchors all excess-return computations.
- **Market Data to Portfolio (Section 4):** The covariance matrix and expected return vector — both derived from daily returns — are the two primary inputs to every portfolio optimization strategy.
- **Market Data to Bubble Detection (Section 5):** Return distributions (skewness, kurtosis) and volatility patterns feed the composite bubble score. The bubble score then flows back into portfolio optimization as a penalty term.
- **Market Data to Technical Analysis (Section 7):** OHLCV data is the direct input for all technical indicators. These indicators in turn become features for the ML prediction pipeline (Section 10).
- **Market Data to ML (Sections 10-11):** Returns, volatility, and volume ratios become ML features. The 6-dimensional feature vector for clustering (return, vol, skew, kurtosis, Sharpe, drawdown) is entirely computed from this module's outputs.

2.1 Data Acquisition

The primary data fetch function downloads adjusted close prices and volume data for all selected tickers simultaneously. MultiIndex column handling ensures correct data extraction when multiple tickers are requested.

Code: `fetch_market_data()` Pattern

```
@st.cache_data(ttl=3600)
def fetch_market_data(tickers, start, end):
    raw = yf.download(tickers, start=start, end=end,
                      auto_adjust=True, group_by='ticker')
    if isinstance(raw.columns, pd.MultiIndex):
```

```

prices = raw.xs('Close', axis=1, level=1)
else:
    prices = raw[['Close']].rename(columns={'Close': tickers[0]})
returns = prices.pct_change().dropna()
return prices, returns

```

2.2 Risk-Free Rate

The risk-free rate is dynamically fetched from the 3-month US Treasury Bill yield (ticker ^IRX) via yfinance. This rate anchors all excess-return calculations including the Sharpe ratio, Sortino ratio, CAPM, and portfolio optimization targets.

2.3 Performance Metrics

QuantLab computes a comprehensive set of risk-adjusted performance metrics for each asset. All calculations assume 252 trading days per year for annualization.

Metric	Formula	Interpretation
Annualized Return	$R_{\text{ann}} = \text{mean}(r_{\text{daily}}) \times 252$	Expected yearly return
Annualized Volatility	$\sigma_{\text{ann}} = \text{std}(r_{\text{daily}}) \times \sqrt{252}$	Yearly risk measure
Sharpe Ratio	$S = (R_{\text{ann}} - R_f) / \sigma_{\text{ann}}$	Risk-adjusted excess return
Maximum Drawdown	$\text{MDD} = \max(1 - P_t / \max(P_{0:t}))$	Worst peak-to-trough decline
Sortino Ratio	$S_o = (R_{\text{ann}} - R_f) / \sigma_{\text{downside}}$	Downside risk-adjusted return
Calmar Ratio	$C = R_{\text{ann}} / \text{MDD} $	Return per unit max drawdown

Annualized Return

The annualized return scales the mean daily arithmetic return to a yearly figure:

$$R_{\text{ann}} = \text{mean}(r_{\text{daily}}) \times 252$$

where $r_{\text{daily}} = (P_t - P_{t-1}) / P_{t-1}$ represents simple daily percentage returns.

Annualized Volatility

$$\sigma_{\text{ann}} = \text{std}(r_{\text{daily}}) \times \sqrt{252}$$

Volatility is computed as the sample standard deviation of daily returns, scaled by the square root of trading days. This assumes i.i.d. returns — an approximation that may understate risk during volatility clustering regimes.

Sharpe Ratio

$$S = (R_{\text{ann}} - R_f) / \sigma_{\text{ann}}$$

The Sharpe ratio measures excess return per unit of total volatility. A ratio above 1.0 is generally considered good; above 2.0 is excellent. The risk-free rate R_f is sourced from the 3-month T-bill yield.

Maximum Drawdown

$$\text{MDD} = \max(1 - P_t / \max(P_0, \dots, P_t))$$

Maximum drawdown captures the largest peak-to-trough decline over the entire period. It is a critical risk metric for institutional investors as it quantifies the worst-case loss scenario an investor would have experienced.

Sortino Ratio

$$S_o = (R_{\text{ann}} - R_f) / \sigma_{\text{downside}}$$

The Sortino ratio refines the Sharpe ratio by penalizing only downside volatility. σ_{downside} is computed from negative returns only, providing a more nuanced risk-adjusted measure that ignores upside variance.

Calmar Ratio

$$C = R_{\text{ann}} / |\text{MDD}|$$

The Calmar ratio divides annualized return by the absolute value of maximum drawdown, offering a return-per-unit-of-drawdown-risk perspective.

3. Valuation Models

QuantLab implements five foundational valuation frameworks used in institutional equity research. These models estimate the intrinsic value of assets by analyzing risk premiums, cost of capital, and discounted future cash flows.

Purpose & Role in the Investment Workflow

Valuation models answer the fundamental question: is this asset overpriced or underpriced relative to its risk? For portfolio managers, these estimates provide the rational basis for position sizing and conviction levels. CAPM establishes the baseline expected return for a given level of systematic risk; WACC translates that into a discount rate for enterprise valuation; DCF converts future cash flow projections into a present-value intrinsic price. The Fama-French and APT models extend this framework to capture additional risk dimensions (size, value, momentum) that CAPM alone misses. Together, these five models provide multiple independent valuation perspectives — convergence among them strengthens conviction, while divergence flags assets requiring deeper fundamental analysis.

Cross-Module Interactions

- **Valuation from Market Data (Section 2):** Beta is computed from the returns covariance between each asset and the S&P 500 benchmark. The risk-free rate from Section 2 anchors all expected return calculations.
- **CAPM to WACC to DCF (Internal Chain):** The five valuation models form an internal dependency chain. CAPM produces the cost of equity (R_e), which flows into WACC as one of two capital components. WACC then becomes the discount rate for the DCF model. Errors in beta estimation therefore propagate multiplicatively through the entire valuation chain.
- **Valuation to Portfolio (Section 4):** The expected returns from CAPM/Fama-French/APT can serve as the mean return vector (μ) for portfolio optimization, providing a fundamentals-driven alternative to historical mean returns.
- **Valuation to Monte Carlo (Section 6):** DCF-derived intrinsic values inform the fundamentalist agent's target price (Θ) in the behavioral Monte Carlo simulation, determining where mean-reverting demand pressure originates.
- **Valuation to Bubble Detection (Section 5):** When market prices significantly exceed DCF intrinsic value, this divergence contributes to bubble score elevation. The Metcalfe MMV ratio provides a parallel, network-based valuation benchmark.

3.1 CAPM (Capital Asset Pricing Model)

The Capital Asset Pricing Model establishes a linear relationship between the expected return of an asset and its systematic risk (beta) relative to the market portfolio.

$$E(R_i) = R_f + \beta_{i,m} \times (E(R_m) - R_f)$$

where:

- $E(R_i)$ = expected return on asset i
- R_f = risk-free rate (3-month T-bill)
- β_i = systematic risk coefficient
- $E(R_m) - R_f$ = market risk premium (5.7%)

Beta Calculation

$$\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$$

Beta is computed using the covariance of the asset's returns with the market returns (S&P 500 proxy) divided by the variance of market returns. QuantLab uses `np.cov()` for this calculation.

Code: `calculate_beta()`

```
def calculate_beta(asset_returns, market_returns):
    cov_matrix = np.cov(asset_returns, market_returns)
    beta = cov_matrix[0, 1] / cov_matrix[1, 1]
    return beta

# Market premium used: 5.7%
expected_return = risk_free + beta * 0.057
```

Institutional Relevance: CAPM provides the baseline cost-of-equity estimate used in WACC calculations and as a performance benchmark. While empirical evidence shows limitations (e.g., low-beta anomaly), it remains the foundation for most institutional valuation frameworks.

3.2 WACC (Weighted Average Cost of Capital)

WACC represents the blended cost of a firm's financing sources, weighted by their proportion in the capital structure. It serves as the discount rate for DCF valuations.

$$\text{WACC} = (E/V) \times R_e + (D/V) \times R_d \times (1 - T_c)$$

where:

- E = market capitalization (equity value)
- D = total debt (from balance sheet)
- $V = E + D$ = total enterprise value
- R_e = cost of equity (from CAPM)
- R_d = cost of debt = interest expense / total debt
- T_c = corporate tax rate (assumed 21%)

The cost of equity R_e flows directly from the CAPM calculation. The cost of debt is derived from the company's interest expense relative to total outstanding debt, with the tax shield $(1 - T_c)$ reflecting the deductibility of interest payments.

Institutional Relevance: WACC is the cornerstone discount rate for enterprise valuation. Errors in WACC estimation propagate multiplicatively through DCF models, making accurate beta and capital structure estimation critical.

3.3 DCF (Discounted Cash Flow)

The DCF model projects future free cash flows and discounts them to present value using the WACC. QuantLab implements a 5-year projection with a terminal value calculation.

Free Cash Flow Projection

$$FCF_t = FCF_0 \times (1 + g)^t$$

where $g = 5\%$ is the assumed growth rate and $t = 1, 2, \dots, 5$ is the projection year.

Present Value of Cash Flows

$$PV_t = FCF_t / (1 + WACC)^t$$

Terminal Value (Gordon Growth Model)

$$TV = FCF_5 \times (1 + g_{\text{terminal}}) / (WACC - g_{\text{terminal}})$$

The terminal growth rate $g_{\text{terminal}} = 2\%$ represents perpetual growth beyond the explicit forecast period, typically aligned with long-run GDP or inflation expectations.

Enterprise Value

$$EV = \text{sum}(PV_1 \dots PV_5) + PV(TV)$$

Enterprise value is the sum of discounted projected cash flows plus the present value of the terminal value. Equity value is then $EV - \text{Net Debt}$, and the intrinsic share price is $\text{Equity Value} / \text{Shares Outstanding}$.

Parameter	Value	Source
Growth Rate (g)	5%	Analyst assumption
Terminal Growth (g_terminal)	2%	Long-run GDP proxy
Projection Horizon	5 years	Standard institutional practice
Discount Rate	WACC	Computed from CAPM + capital structure
Tax Rate	21%	US statutory corporate rate

3.4 Fama-French Three-Factor Model

The Fama-French three-factor model extends CAPM by adding size (SMB) and value (HML) risk premiums to explain cross-sectional return variation.

$$E(R_i) = R_f + \text{beta}_m \times \text{MRP} + \text{beta}_{\text{smb}} \times \text{SMB} + \text{beta}_{\text{hml}} \times \text{HML}$$

where:

- MRP = Market Risk Premium = 5.7%
- SMB = Small Minus Big = 2% (size premium)
- HML = High Minus Low = 4% (value premium)
- $\text{beta}_{\text{smb}}, \text{beta}_{\text{hml}}$ = factor loadings inferred from market beta

QuantLab infers factor loadings from market beta: high-beta assets receive a growth tilt (negative HML loading), while low-beta assets receive a value tilt (positive HML loading). This heuristic approximation avoids requiring access to the Fama-French factor data files.

Institutional Relevance: The Fama-French model provides a more complete picture of expected returns for fund managers running factor-tilted portfolios. The SMB and HML premiums have been extensively documented across global equity markets.

3.5 APT (Arbitrage Pricing Theory)

APT generalizes CAPM by allowing multiple systematic risk factors without specifying the market portfolio. QuantLab implements a three-factor APT model.

$$E(R_i) = R_f + \sum(\text{beta}_k \times \text{lambda}_k) \text{ for } k = 1..K$$

Factor (k)	Risk Premium (lambda)	Description
Market	5.7%	Broad equity market risk
Volatility	Calibrated	VIX-linked volatility factor
Momentum	Calibrated	12-1 month momentum factor

Factor sensitivities are calibrated from historical price data. The APT framework's flexibility allows institutional users to incorporate macro factors, sector exposures, or proprietary signal betas.

4. Portfolio Optimization

QuantLab implements eight portfolio optimization strategies, all built on the mean-variance framework and solved via `scipy.optimize.minimize` with the SLSQP algorithm. Each strategy includes an optional bubble-aware penalty mechanism that dynamically adjusts weights based on bubble detection scores.

Purpose & Role in the Investment Workflow

Portfolio optimization is where all upstream analytics converge into actionable allocation decisions. While Sections 2-3 answer 'what is each asset worth and how risky is it?', this module answers the critical question: 'how much capital should I allocate to each asset?' The eight strategies span the full spectrum of institutional needs — from the classic Max Sharpe tangency portfolio favored by active managers, to Risk Parity used by macro hedge funds (e.g., Bridgewater's All Weather), to Kelly Criterion used by systematic traders maximizing geometric growth. The bubble-aware penalty mechanism is QuantLab's unique contribution: it dynamically integrates the Bubble Detection module's output directly into the optimization objective, ensuring that position sizing accounts for overvaluation risk in real time.

Cross-Module Interactions

- **Portfolio from Market Data (Section 2):** The covariance matrix (Σ) and expected return vector (μ) are the two fundamental inputs. Annualized returns define the 'reward' axis; the covariance matrix defines the 'risk' surface over which the optimizer searches.
- **Portfolio from Bubble Detection (Section 5):** The composite bubble score for each asset is injected as a quadratic penalty in the objective function. High bubble scores increase the cost of concentrating weight in overvalued assets, causing the optimizer to redistribute toward fundamentally sound holdings.
- **Portfolio from Valuation (Section 3):** Black-Litterman uses market-implied equilibrium returns (derived from CAPM) as prior expectations, then blends them with investor views. The risk aversion coefficient (λ) links CAPM beta to equilibrium allocation weights.
- **Portfolio to Monte Carlo (Section 6):** Optimized portfolio weights can be forward-tested using Monte Carlo simulation to estimate the distribution of future portfolio returns under different market regimes.
- **Portfolio to Risk Dashboard (Section 9):** Portfolio assets feed into the cross-asset correlation matrix, revealing whether the optimized portfolio's diversification holds during stress events or breaks down due to correlation spikes.
- **Portfolio to Export (Section 13):** All eight strategy weights, efficient frontier data, and performance comparisons are exported to PDF slides and Excel sheets for client reporting and investment committee presentations.

4.1 Mean-Variance Framework

The foundational Markowitz framework maximizes risk-adjusted returns subject to full investment and long-only constraints.

Objective Function

$$\max S = (w' \times \mu - R_f) / \sqrt{w' \times \Sigma \times w}$$

Constraints

$$\text{subject to: } \sum(w_i) = 1, w_i \geq 0 \text{ for all } i$$

The covariance matrix is annualized: $\Sigma = \text{Cov}(R) \times 252$. Expected returns are annualized arithmetic means. The optimizer uses Sequential Least Squares Programming (SLSQP) which handles both equality and inequality constraints efficiently.

Code: Mean-Variance Optimization Core

```
from scipy.optimize import minimize

def optimize_portfolio(mu, cov, rf):
    n = len(mu)
    w0 = np.ones(n) / n # equal weight initial guess
    bounds = [(0, 1)] * n
    constraints = [{'type': 'eq', 'fun': lambda w: np.sum(w) - 1}]

    def neg_sharpe(w):
        ret = w @ mu
        vol = np.sqrt(w @ cov @ w)
        return -(ret - rf) / vol

    result = minimize(neg_sharpe, w0, method='SLSQP',
                      bounds=bounds, constraints=constraints)
    return result.x
```

4.2 Maximum Sharpe Ratio

Maximizes the Sharpe ratio to find the tangency portfolio on the efficient frontier. The bubble-aware variant adds a quadratic penalty term that penalizes allocation to assets with elevated bubble scores.

Bubble-Aware Adjustment

$$S_{\text{adjusted}} = S - \text{penalty_factor} \times \sum(\text{bubble_score}_i \times w_i^2)$$

The penalty factor is user-configurable (default 0.5). Higher bubble scores exponentially increase the cost of concentrated positions in overvalued assets, causing the optimizer to redistribute weight toward fundamentally sound holdings.

4.3 Minimum Variance

The minimum variance portfolio minimizes total portfolio volatility regardless of expected returns.

$$\min w' \times \Sigma \times w$$

Bubble Penalty

$$\text{Objective} = w' \times \text{Sigma} \times w + \text{penalty} \times \sum(\text{score}_i \times w_i^2)$$

This portfolio is popular among risk-averse investors and has shown empirical outperformance in bear markets. The bubble penalty shifts weight away from potentially overvalued low-volatility assets.

4.4 Risk Parity

Risk parity targets equal risk contribution from each portfolio asset, providing more balanced risk diversification than equal weighting.

Marginal Risk Contribution

$$MC_i = (\text{Sigma} \times w)_i / \text{sigma}_p$$

Risk Contribution

$$RC_i = w_i \times MC_i$$

Optimization Target

$$\min \sum((RC_i - \text{sigma}_p / N)^2)$$

Risk parity portfolios typically overweight bonds and underweight equities compared to market-cap-weighted benchmarks. This approach became popularized by Bridgewater Associates' All Weather strategy.

4.5 Minimum CVaR

Conditional Value-at-Risk (CVaR) measures the expected loss beyond the VaR threshold, providing a coherent risk measure that captures tail risk.

$$CVaR_{\alpha} = E[\text{Loss} \mid \text{Loss} > VaR_{\alpha}]$$

QuantLab computes a CVaR-based covariance matrix from tail returns (returns below the alpha quantile). Positive returns are zeroed before computing the semi-covariance matrix, focusing the optimization entirely on downside risk.

4.6 Maximum Diversification

Maximizes the diversification ratio — the ratio of weighted average individual volatilities to portfolio volatility.

$$DR = (w' \times \text{sigma}_{\text{individual}}) / \text{sigma}_{\text{portfolio}}$$

A higher diversification ratio indicates that correlation effects are reducing portfolio risk beyond what individual asset volatilities would suggest. The maximum diversification portfolio seeks the maximum benefit from imperfect correlations.

4.7 Kelly Criterion

The Kelly criterion determines the optimal bet size to maximize long-run geometric growth rate. QuantLab adapts it for multi-asset portfolio allocation.

$$w^* = \text{Sigma}^{-1} \times (\mu - R_f)$$

Full Kelly weights can be extreme, so QuantLab applies leverage limits and converts to long-only allocations. If the covariance matrix is singular (e.g., highly correlated assets), the Moore-Penrose pseudo-inverse (`np.linalg.pinv`) is used.

4.8 Black-Litterman

The Black-Litterman model combines market-implied equilibrium returns with subjective investor views through Bayesian updating, producing more stable and intuitive portfolio weights.

Market-Implied Returns

$$P_i = \text{lambda} \times \text{Sigma} \times w_{\text{mkt}}$$

where `lambda` is the risk aversion coefficient.

Posterior Expected Returns

Investor views are incorporated via a pick matrix `P` (identity matrix — one view per asset) with uncertainty `tau` = 0.05. The posterior return vector blends equilibrium and views weighted by their relative certainty.

Institutional Relevance: Black-Litterman addresses well-known instabilities in mean-variance optimization — small changes in expected returns can produce wildly different portfolios. By anchoring to market equilibrium, the model produces more robust allocations.

5. Bubble Detection

QuantLab's bubble detection module implements four complementary approaches to identify asset price bubbles. The system produces a composite score that integrates network valuation, long-memory analysis, fat-tail statistics, and volatility clustering into a single actionable metric.

Purpose & Role in the Investment Workflow

Bubble detection serves as QuantLab's early warning system. While traditional valuation models can identify overpricing, they cannot distinguish between rationally high valuations (driven by genuine growth expectations) and irrational price bubbles (driven by speculative feedback loops). This module addresses that gap by analyzing the statistical fingerprint of bubble dynamics: persistent momentum (long memory), fat-tailed return distributions, volatility clustering, and deviation from network-fundamental value. The composite bubble score is not merely informational — it actively modifies portfolio optimization behavior, making this module a bridge between diagnosis and action. For institutional risk managers, this provides a systematic, model-driven alternative to subjective bubble assessments.

Cross-Module Interactions

- **Bubble Detection from Market Data (Section 2):** The module consumes daily returns, price series, and volume data. Return distribution statistics (skewness, kurtosis) from Section 2 directly feed the composite score calculation.
- **Bubble Detection to Portfolio (Section 4):** This is the most critical cross-module link. The composite bubble score for each asset is injected as a quadratic penalty into all eight portfolio optimization strategies. A high bubble score (e.g., 0.8) causes the optimizer to dramatically reduce weight in that asset, effectively implementing an automated overvaluation hedge.
- **Bubble Detection to Monte Carlo (Section 6):** Bubble regime classification (Normal/Bubble/Crash) informs the Monte Carlo simulation's regime detection framework. When the bubble detector flags 'Bubble Formation', the Monte Carlo module can estimate the probability distribution of subsequent crash scenarios.
- **Metcalf's Law to Valuation (Section 3):** The Metcalfe network valuation provides an alternative intrinsic value estimate for technology and crypto assets, complementing the DCF-based valuation from Section 3. Divergence between DCF and Metcalfe values can itself be a signal.
- **Bubble Detection to Sentiment (Section 12):** Elevated bubble scores combined with strongly bullish sentiment create a contrarian signal — extreme optimism during detected bubble conditions historically precedes corrections.

5.1 Metcalfe's Law (Advanced Network Valuation)

For network-effect assets (technology stocks, cryptocurrencies), QuantLab applies a generalized Metcalfe's law to estimate fundamental network value.

Network Value Model

$$V = A_0 \times (n \times (n - 1) / 2)^{\text{gamma}}$$

where:

- $A_0 = 6.053 \times 10^{-7}$ (scaling constant from Bitcoin research)
- $\text{gamma} = 1.166$ (network elasticity parameter)
- n = network user count, proxied by 30-day smoothed rolling mean of trading volume

MMV Ratio (Metcalf Market Value Ratio)

$$\text{MMV} = \text{Market_Price} / \text{Network_Value}$$

MMV Range	Classification	Interpretation
> 1.25	Extreme Bubble	Price far exceeds network fundamentals
1.15 - 1.25	Bubble Formation	Early overvaluation signals
0.95 - 1.05	Fair Value	Price aligns with network value
0.85 - 0.95	Mildly Undervalued	Potential buying opportunity
< 0.85	Significantly Undervalued	Strong fundamental support

5.2 Long Memory (GPH Estimator)

The Geweke-Porter-Hudak (GPH) estimator detects long-range dependence in asset prices — a characteristic feature of bubble dynamics where past price movements have persistent effects on future prices.

Fractional Differencing Parameter

The GPH method estimates the fractional differencing parameter d by regressing the log periodogram on log frequencies:

$$\log(I(\omega_j)) = c - d \times \log(4 \times \sin^2(\omega_j / 2)) + \epsilon_j$$

$$d = -\beta_1 / 2$$

where β_1 is the OLS slope coefficient. The bandwidth is set to $m = n^{0.5}$ where n is the sample size.

When $d > 0.5$, the series is non-stationary with strong persistence — a hallmark of bubble behavior where mean-reversion is absent.

5.3 Composite Bubble Score

QuantLab combines four sub-indicators into a single composite bubble score on a $[0, 1]$ scale:

Component	Weight Range	Calculation Method
MMV Deviation	0 - 0.4	Sigmoid scaling of (MMV - 1)
Long Memory d	0 - 0.3	d x 0.3 (capped)
Excess Kurtosis	0 - 0.2	Fat tails = elevated bubble risk
Volatility Clustering	0 - 0.1	Ljung-Box test on squared returns

The composite score is used directly in the bubble-aware portfolio optimization strategies, where it modulates the quadratic penalty applied to each asset's weight.

5.4 Statistical Tests

QuantLab performs several statistical tests to characterize return distributions:

- **Skewness:** Computed via `scipy.stats.skew()`. Negative skewness indicates left-tail risk; positive skewness may indicate speculative behavior.
- **Excess Kurtosis:** Via `scipy.stats.kurtosis()`. Values above 3 indicate heavier tails than normal — increased probability of extreme events.
- **Ljung-Box Test:** Applied to squared returns using `statsmodels.acorr_ljungbox()` with 10 lags. Significant p-values indicate ARCH effects (volatility clustering), a common feature of bubble periods.

6. Monte Carlo Simulation — Behavioral Agent Model

QuantLab implements a heterogeneous agent-based Monte Carlo simulation that models market dynamics as the interaction between fundamentalist and speculator agent populations. This approach captures regime switching, bubble formation, and crash dynamics that standard geometric Brownian motion cannot reproduce.

Purpose & Role in the Investment Workflow

Monte Carlo simulation is the forward-looking engine of QuantLab. While all other modules analyze historical data, this module projects into the future by simulating thousands of possible price paths. The behavioral agent model is a deliberate departure from standard geometric Brownian motion (GBM): GBM assumes returns are normally distributed and independent, but real markets exhibit fat tails, volatility clustering, and regime switches. By modeling markets as the interaction between fundamentalist (mean-reverting) and speculator (trend-following) agents, QuantLab can reproduce realistic crash dynamics, bubble inflation, and regime transitions. For hedge funds and institutional risk desks, this produces more realistic Value-at-Risk estimates and stress-test scenarios than parametric VaR models.

Cross-Module Interactions

- **Monte Carlo from Market Data (Section 2):** Historical volatility, autocorrelation, and distribution parameters calibrate the agent model. The starting price for simulations is the last observed close price.
- **Monte Carlo from Valuation (Section 3):** The fundamentalist agent's target price (Theta) is informed by intrinsic value estimates. The drift rate of the intrinsic value process is calibrated from historical return patterns.
- **Monte Carlo from Bubble Detection (Section 5):** The fundamentalist weight parameter (rho) is inversely related to bubble score — during detected bubbles, speculator demand dominates, amplifying trend-following dynamics in the simulation.
- **Monte Carlo to Portfolio (Section 4):** Simulated forward return distributions can stress-test portfolio allocations. Users can evaluate how their optimized portfolio would perform across the range of Monte Carlo scenarios.
- **Monte Carlo to Risk Dashboard (Section 9):** Regime transition probabilities (Normal to Bubble to Crash) from the simulation complement the real-time risk score, providing a forward-looking risk dimension.

6.1 Agent-Based Model Design

The model features two agent types with competing demand functions that drive price evolution:

Price Evolution

$$P_t = P_{t-1} \times \exp(\kappa \times D / P + Z)$$

where $\kappa = 0.1$ is the price impact coefficient, D is aggregate excess demand, and Z is an exogenous noise term.

Aggregate Excess Demand

$$D = \rho \times D_f + (1 - \rho) \times D_s$$

Fundamentalist Demand

$$D_f = \beta_f \times (\Theta_t - P_{t-1})$$

Fundamentalists are mean-reverting agents who buy when price is below intrinsic value Θ_t and sell when above. Their demand is proportional to the price-value gap.

Speculator Demand

$$D_s = \beta_s \times (P_{t-1} - \mu_s) / (\text{risk_aversion} \times \nu_s)$$

Speculators are trend-following agents whose demand is driven by recent price momentum relative to their exponentially weighted moving average μ_s , scaled by their risk tolerance.

6.2 Parameter Calibration

Model parameters are calibrated from historical market data to match observed statistical properties of each asset:

Parameter	Calibration Rule	Typical Range
ρ (fundamentalist weight)	From β , clipped to $[0.2, 0.9]$	0.2 - 0.9
β_s (speculator sensitivity)	$\text{lag-1 autocorr} \times 0.3 + 0.02$	0.02 - 0.10
δ (EWMA decay)	$0.99 - (\text{ann_vol} \times 0.05)$, clipped	0.92 - 0.995
γ (network effect)	1.5 for tech/crypto, 1.0 otherwise	1.0 - 1.5
κ (price impact)	Fixed constant	0.1

6.3 Intrinsic Value & Regime Detection

The intrinsic value process follows geometric Brownian motion with calibrated drift:

$$\Theta_t = \Theta_{t-1} \times \exp((\mu_{\theta} - \sigma_{\theta}^2/2) \times dt + \sigma_{\theta} \times dW)$$

Divergence & Regime Classification

$$\text{div} = P_t / \Theta_t$$

Divergence Range	Regime	Code
> 1.15	Bubble	2
0.85 - 1.15	Normal	1
< 0.85	Crash	0

The simulation outputs multiple paths (configurable, default 1000), from which confidence intervals, expected returns, and regime transition probabilities are computed.

7. Technical Analysis

QuantLab computes a full suite of technical indicators using the `ta` library. These indicators are used both for standalone technical analysis and as features in the machine learning prediction pipeline.

Purpose & Role in the Investment Workflow

Technical analysis provides the timing dimension that fundamental analysis lacks. While Valuation (Section 3) tells you what to buy and Portfolio Optimization (Section 4) tells you how much, technical indicators help answer when to execute. SMA crossovers signal trend changes, RSI identifies overbought/oversold extremes, and Bollinger Bands measure volatility regimes. For systematic traders, these indicators generate rule-based entry/exit signals. For discretionary portfolio managers, they provide confirmation or warning signals overlaid on fundamental views. Critically, these indicators also serve a dual role as machine learning features (Section 10), where they capture momentum, mean-reversion, and volatility dynamics that improve forward return predictions.

Cross-Module Interactions

- **Technicals from Market Data (Section 2):** All technical indicators are computed from OHLCV price and volume data. The quality and length of the historical data window directly affects indicator reliability — a 200-day SMA requires at least 200 trading days of data.
- **Technicals to ML Predictions (Section 10):** RSI, SMA ratios (Price/SMA_20, Price/SMA_50, Price/SMA_200), and volume ratios are core features in the ML feature vector. The ML module depends on these indicators being pre-computed and available in the DataFrame.
- **Technicals to Bubble Detection (Section 5):** Persistent RSI readings above 70 (overbought) combined with Bollinger Band expansion provide supplementary confirmation of bubble-like price behavior detected by the statistical methods in Section 5.
- **Technicals to Sentiment (Section 12):** Divergence between technical signals (e.g., bearish MACD crossover) and sentiment signals (e.g., bullish headlines) creates actionable contrarian setups that sophisticated traders monitor.
- **Technicals to Export (Section 13):** Technical indicator time series are exported to dedicated Excel sheets and visualized as overlay charts in PDF/slide reports.

7.1 Moving Averages

Simple Moving Average (SMA)

$$SMA_n = (1/n) \times \sum(P_{t-i}) \text{ for } i = 0 \text{ to } n-1$$

Computed for windows of 20, 50, and 200 days. SMA crossovers (e.g., 50-day crossing above 200-day = 'golden cross') generate classic trend signals.

Exponential Moving Average (EMA)

$$EMA_t = \alpha \times P_t + (1 - \alpha) \times EMA_{t-1}$$

$$\alpha = 2 / (n + 1)$$

EMA places greater weight on recent prices, making it more responsive to new information than SMA. Standard periods: 12-day and 26-day for MACD computation.

7.2 MACD (Moving Average Convergence Divergence)

$$MACD = EMA_{12} - EMA_{26}$$

$$Signal = EMA_9(MACD)$$

MACD crossovers with the signal line generate buy/sell signals. The MACD histogram (MACD - Signal) visualizes momentum strength. Implemented via `ta.trend.MACD()`.

7.3 RSI (Relative Strength Index)

$$RSI = 100 - 100 / (1 + RS)$$

$$RS = avg_gain_{14} / avg_loss_{14}$$

RSI ranges from 0 to 100. Readings above 70 indicate overbought conditions; below 30 indicates oversold. Implemented via `ta.momentum.RSIIndicator()`.

7.4 Bollinger Bands

$$Upper = SMA_{20} + 2 \times \sigma_{20}$$

$$Lower = SMA_{20} - 2 \times \sigma_{20}$$

Bollinger Bands define a dynamic volatility envelope around the 20-day SMA. Price touching the upper band suggests overbought; touching the lower band suggests oversold. Band squeeze (narrowing) often precedes volatility expansion.

Code: `TechnicalIndicators.calculate_all()`

```
class TechnicalIndicators:
    @staticmethod
    def calculate_all(df):
        close = df['Close']
        # Moving Averages
        df['SMA_20'] = ta.trend.sma_indicator(close, 20)
        df['SMA_50'] = ta.trend.sma_indicator(close, 50)
        df['EMA_12'] = ta.trend.ema_indicator(close, 12)
        # MACD
        macd = ta.trend.MACD(close)
        df['MACD'] = macd.macd()
        df['MACD_Signal'] = macd.macd_signal()
        # RSI
        df['RSI'] = ta.momentum.RSIIndicator(close, 14).rsi()
```

```
# Bollinger Bands
bb = ta.volatility.BollingerBands(close, 20, 2)
df['BB_Upper'] = bb.bollinger_hband()
df['BB_Lower'] = bb.bollinger_lband()
return df
```

8. Options Pricing

QuantLab implements the Black-Scholes option pricing framework with full Greeks computation and options strategy payoff analysis. The module also supports implied volatility surface visualization from live market data.

Purpose & Role in the Investment Workflow

Options pricing extends QuantLab beyond directional equity analysis into the derivatives space. For institutional desks, options serve three primary functions: hedging existing portfolio risk (e.g., protective puts), generating income (e.g., covered calls), and expressing non-directional views (e.g., straddles for volatility plays). The Black-Scholes calculator provides theoretical fair value for European options, the Greeks quantify risk sensitivities for position management, and the implied volatility surface reveals the market's collective expectation of future volatility across strikes and expirations. This module enables traders to identify mispricings between theoretical and market prices, structure multi-leg strategies, and manage portfolio-level Greeks exposure.

Cross-Module Interactions

- **Options from Market Data (Section 2):** The current stock price (S) and historical volatility (sigma) are required inputs for Black-Scholes pricing. The risk-free rate (r) from Section 2's Treasury yield fetch is used as the continuous discounting rate.
- **Options from Risk Dashboard (Section 9):** VIX levels from the Risk Dashboard provide context for implied volatility analysis — when the IV surface shows elevated skew, cross-referencing with VIX confirms whether this reflects systemic fear or asset-specific risk.
- **Options to Portfolio (Section 4):** Options strategies (protective puts, collars) can be used to hedge optimized portfolio positions. Delta-neutral hedging using Greeks from this module allows portfolio managers to isolate alpha from market direction.
- **IV Surface to Bubble Detection (Section 5):** Extreme implied volatility skew (very high put IV relative to call IV) signals market participants paying a premium for downside protection, which corroborates bubble-phase risk detected by Section 5.
- **Options to Monte Carlo (Section 6):** The Monte Carlo simulated price paths can be used to value path-dependent options (barrier options, Asian options) that Black-Scholes cannot handle in closed form.

8.1 Black-Scholes Model

The Black-Scholes model provides closed-form solutions for European option pricing under the assumptions of log-normal price dynamics, constant volatility, and no dividends.

Call Option Price

$$C = S \times N(d_1) - K \times e^{-rT} \times N(d_2)$$

Put Option Price

$$P = K \times e^{-rT} \times N(-d_2) - S \times N(-d_1)$$

d_1 and d_2 Parameters

$$d_1 = [\ln(S/K) + (r + \sigma^2/2) \times T] / (\sigma \times \sqrt{T})$$

$$d_2 = d_1 - \sigma \times \sqrt{T}$$

where:

- S = current stock price
- K = strike price
- r = risk-free interest rate
- T = time to expiration (in years)
- σ = annualized volatility
- $N(x)$ = cumulative standard normal distribution function
- $n(x)$ = standard normal probability density function

8.2 Greeks

The Greeks measure the sensitivity of option prices to changes in underlying parameters. QuantLab computes all five primary Greeks for both calls and puts.

Greek	Call Formula	Interpretation
Delta	$N(d_1)$	Price sensitivity to underlying
Gamma	$n(d_1) / (S \times \sigma \times \sqrt{T})$	Delta sensitivity to underlying
Theta	$[-S \times n(d_1) \times \sigma / (2 \times \sqrt{T}) - r \times K \times e^{-rT} \times N(d_2)] / 365$	Time decay per day
Vega	$S \times n(d_1) \times \sqrt{T} / 100$	Sensitivity to volatility (per 1%)
Rho	$K \times T \times e^{-rT} \times N(d_2) / 100$	Sensitivity to interest rate (per 1%)

Put Greeks: Delta(put) = $N(d_1) - 1$, Theta and Rho use $N(-d_2)$ variants. Gamma and Vega are identical for calls and puts.

8.3 Options Strategies

QuantLab supports payoff analysis for multiple options strategies:

Strategy	Construction	Max Profit / Max Loss
Long Call	Buy call at strike K	Unlimited / Premium paid
Long Put	Buy put at strike K	$(K - \text{Premium})$ / Premium paid
Bull Call Spread	Buy call K_1 , sell call K_2 ($K_2 > K_1$)	$(K_2 - K_1 - \text{net prem})$ / Net premium

Strategy	Construction	Max Profit / Max Loss
Bear Put Spread	Buy put K2, sell put K1 ($K2 > K1$)	$(K2 - K1 - \text{net prem}) / \text{Net premium}$
Straddle	Buy call + put at same K	Unlimited / Total premium
Iron Condor	Bull put spread + bear call spread	Net credit / $(\text{Width} - \text{credit})$

Code: options_payoff()

```
def options_payoff(S_range, K, premium, option_type='call'):
    if option_type == 'call':
        payoff = np.maximum(S_range - K, 0) - premium
    else:
        payoff = np.maximum(K - S_range, 0) - premium
    return payoff

def bull_call_spread(S_range, K1, K2, net_premium):
    return (np.maximum(S_range - K1, 0)
            - np.maximum(S_range - K2, 0) - net_premium)
```

8.4 Implied Volatility Surface

QuantLab generates 3D implied volatility surfaces plotting IV against strike price and time to expiration. Options chain data is fetched via `yf.Ticker(symbol).option_chain(date)` for each available expiration date.

The volatility smile/skew pattern visible in the surface reveals market expectations for tail risk — typically, out-of-the-money puts carry higher IV than calls (negative skew), reflecting demand for downside protection.

9. Risk & Geopolitics Dashboard

The Risk & Geopolitics Dashboard provides a real-time macro-risk monitoring panel that aggregates multiple market indicators into a composite risk score. This module is designed for institutional portfolio managers who need to assess systemic risk exposure.

Purpose & Role in the Investment Workflow

While other modules analyze individual assets or portfolios, the Risk Dashboard monitors the macro environment in which all investments operate. A portfolio may be perfectly optimized (Section 4), fundamentally sound (Section 3), and technically well-timed (Section 7) — but if systemic risk spikes (e.g., yield curve inversion, VIX surge, gold flight-to-safety), all risk assets can decline simultaneously, overwhelming micro-level analysis. This module provides the macro overlay that institutional risk committees require: a single composite score (0-100) that summarizes systemic risk, plus granular indicator panels for volatility, rates, currencies, and safe-haven flows. The cross-asset correlation matrix is particularly critical because correlation breakdown during stress events can cause diversified portfolios to suffer unexpected concentrated losses.

Cross-Module Interactions

- **Risk Dashboard from Market Data (Section 2):** Macro indicators (VIX, Treasury yields, Gold, Oil, DXY) are fetched using the same yfinance data pipeline, ensuring consistent timestamps with portfolio asset data.
- **Risk Dashboard to Portfolio (Section 4):** The cross-asset correlation matrix reveals whether portfolio diversification assumptions hold. During stress periods, correlations spike toward 1.0, meaning a Risk Parity portfolio may not deliver the risk balance it was designed for. This module provides the monitoring layer to detect such regime shifts.
- **Risk Dashboard to Monte Carlo (Section 6):** The current regime classification (Green/Yellow/Red) contextualizes Monte Carlo outputs. Simulated scenarios under a Red regime carry different probability weightings than those run during Green conditions.
- **Risk Dashboard to Options (Section 8):** VIX levels directly affect options pricing — elevated VIX means higher implied volatilities and more expensive hedges. The Risk Dashboard helps traders decide whether current hedge costs are justified by systemic risk levels.
- **Risk Dashboard to Bubble Detection (Section 5):** Macro risk indicators provide a macro-level sanity check on bubble signals. A high bubble score during low macro risk (Green) may indicate asset-specific froth; a high bubble score during high macro risk (Red) suggests broader systemic fragility.
- **Yield Curve to Valuation (Section 3):** Treasury yield data (10Y, 5Y, 3M) from this dashboard is used in WACC calculations and as the risk-free rate for CAPM. An inverted yield curve simultaneously signals recession risk here and affects discount rates in Section 3.

9.1 Risk Indicators

Indicator	Ticker	Role
CBOE Volatility Index (VIX)	^VIX	Equity market fear gauge
US Dollar Index (DXY)	DX-Y.NYB	USD strength / risk appetite
Gold	GC=F	Safe haven demand proxy
Crude Oil	CL=F	Energy / geopolitical risk
10-Year Treasury Yield	^TNX	Long-term rate expectations
3-Month Treasury Yield	^IRX	Short-term rate / risk-free proxy

9.2 Composite Risk Score [0-100]

QuantLab computes a composite risk score from four equally-weighted components, each contributing up to 25 points:

Component	Weight	Methodology
VIX Level	25 pts	Normalized from 10-50 range (linear scaling)
Yield Curve	25 pts	10Y-3M spread; inversion (negative) = high risk
Gold Return	25 pts	30-day gold return; positive = flight to safety
VIX Regime	25 pts	21d avg / 252d avg ratio; >1 = elevated regime

Score Range	Signal	Action Guidance
0 - 30	Green (Low Risk)	Normal market conditions; standard positioning
30 - 60	Yellow (Moderate Risk)	Heightened vigilance; review hedges
60 - 100	Red (High Risk)	Defensive posture; increase hedges / reduce exposure

9.3 Cross-Asset Correlation Matrix

The dashboard generates a Pearson correlation matrix across VIX, Gold, Oil, DXY, TNX, S&P 500, and all portfolio assets. Correlation regime shifts — such as equities and bonds becoming positively correlated during stress events — are key indicators of systemic risk that can undermine traditional diversification assumptions.

Institutional Relevance: Correlation breakdown monitoring is essential for risk managers. During crises, cross-asset correlations tend toward 1.0 ("correlations go to 1"), which can cause diversified portfolios to suffer unexpected concentrated losses.

10. Machine Learning — Predictive Models

QuantLab integrates three machine learning models for predicting forward asset returns. The ML pipeline follows institutional best practices: chronological train/test splitting, feature standardization, and multiple evaluation metrics.

Purpose & Role in the Investment Workflow

ML Predictions represent QuantLab's quantitative alpha generation layer. While valuation models (Section 3) estimate where price should be based on fundamentals, and technical analysis (Section 7) identifies patterns in historical price action, the ML module combines both into a data-driven forward return forecast. The three-model ensemble (Linear Regression, Random Forest, Gradient Boosting) provides robustness — linear models capture simple factor relationships while tree-based models capture non-linear interactions between features. Feature importance rankings reveal which market signals carry the most predictive power, providing quantitative confirmation (or contradiction) of fundamental hypotheses. For quantitative hedge funds, these predictions can drive systematic trading signals; for discretionary managers, they serve as a 'second opinion' that incorporates dozens of variables simultaneously.

Cross-Module Interactions

- **ML from Market Data (Section 2):** Returns (1d, 5d, 21d) and volatility features are computed from Section 2's return series. The 21-day forward return target variable is derived from future prices in the same dataset.
- **ML from Technical Analysis (Section 7):** RSI, SMA ratios (P/SMA_20, P/SMA_50, P/SMA_200), and volume ratios are core features. Without Section 7's indicator computation, the ML feature vector would lose its momentum and trend signals.
- **ML to Portfolio (Section 4):** ML-predicted forward returns can replace historical mean returns as the expected return vector (μ) in portfolio optimization, providing a forward-looking allocation that adapts to current market conditions.
- **ML to Clustering (Section 11):** Both ML modules share the scikit-learn infrastructure and feature engineering pipeline. Feature importance from Section 10's tree models can inform which features to emphasize in Section 11's clustering feature vector.
- **ML to Export (Section 13):** Model predictions, accuracy metrics (R-squared, RMSE, MAE), and feature importance charts are exported to Excel sheets and PDF report pages for quantitative review.

10.1 Feature Engineering

The feature set is designed to capture momentum, volatility, trend, and volume dynamics:

Feature	Formula / Description	Rationale
Returns (1d, 5d, 21d)	Percentage price changes	Short/medium momentum
Volatility (21d)	Rolling std of daily returns	Risk regime indicator

Feature	Formula / Description	Rationale
SMA Ratio (P/SMA_20)	Price / 20-day SMA	Short-term trend
SMA Ratio (P/SMA_50)	Price / 50-day SMA	Medium-term trend
SMA Ratio (P/SMA_200)	Price / 200-day SMA	Long-term trend
Volume Ratio	Volume / 20-day SMA(Volume)	Unusual activity detection
RSI (14-day)	Relative Strength Index	Overbought/oversold signal

Target Variable: Forward 21-day return (approximately 1 month ahead). This horizon balances signal strength against noise — shorter horizons are dominated by microstructure noise, while longer horizons reduce the effective training sample size.

10.2 Models

Linear Regression

$$y = X \times \text{beta} + \text{epsilon}$$

The baseline model fits a linear relationship between features and forward returns via ordinary least squares. Serves as a benchmark for tree-based model comparison.

Random Forest

An ensemble of 100 decision trees, each trained on a bootstrap sample with maximum depth of 8. Random feature subsampling at each split reduces overfitting.

Hyperparameter	Value	Purpose
n_estimators	100	Number of trees in ensemble
max_depth	8	Limit tree complexity to prevent overfitting
max_features	sqrt(n)	Random feature subsampling per split

Gradient Boosting

Sequential ensemble that builds trees to correct residual errors from previous iterations, typically achieving better accuracy than Random Forest at the cost of increased overfitting risk.

Hyperparameter	Value	Purpose
n_estimators	100	Boosting iterations
learning_rate	0.1	Shrinkage per iteration
max_depth	5	Shallower trees for gradient boosting

10.3 Training Protocol

The training protocol is designed to prevent lookahead bias — the most critical pitfall in financial ML:

- **Chronological Split (80/20):** The first 80% of data (by date) is used for training; the most recent 20% for testing. No random shuffling.
- **Feature Scaling:** `StandardScaler` is fitted on training data only, then applied to both train and test sets.
- **Extended Training Window:** Configurable from 1 to 5 years (default 3 years) to balance signal stability against regime change adaptation.
- **No Cross-Validation:** Time-series cross-validation is not used in the default configuration to maintain simplicity, though it could be added for research purposes.

Evaluation Metrics

Metric	Formula	Ideal Value
R-squared	$1 - SS_{\text{res}} / SS_{\text{tot}}$	Higher is better; >0 beats naive mean
RMSE	$\sqrt{\text{mean}((y - y_{\text{hat}})^2)}$	Lower is better
MAE	$\text{mean}(y - y_{\text{hat}})$	Lower is better; robust to outliers

Feature importance rankings from tree-based models reveal which market signals carry the most predictive power for forward returns. This information is valuable for fundamental analysts seeking quantitative confirmation of their hypotheses.

11. ML Clustering & Regime Detection

QuantLab's clustering module provides unsupervised learning tools for asset classification, market regime identification, anomaly detection, and factor analysis.

Purpose & Role in the Investment Workflow

While ML Predictions (Section 10) use supervised learning to forecast returns, this module uses unsupervised learning to discover hidden structure in the data. Asset clustering reveals natural peer groups that may differ from traditional sector classifications — two companies in different sectors may exhibit similar risk-return profiles and deserve to be treated as substitutes in portfolio construction. Market regime detection identifies whether the market is currently in a Bull, Bear, or Sideways state, which has profound implications for strategy selection: momentum strategies thrive in trending regimes but suffer during reversals. Anomaly detection flags unusual trading days that may indicate institutional order flow, corporate events, or market manipulation. PCA factor analysis reveals the latent risk drivers beneath surface-level asset correlations, answering the question: how many independent sources of risk actually drive this portfolio?

Cross-Module Interactions

- **Clustering from Market Data (Section 2):** The 6-dimensional clustering feature vector (return, volatility, skewness, kurtosis, Sharpe, drawdown) is entirely computed from Section 2's performance metrics. Changes in the analysis date range directly affect cluster assignments.
- **Regime Detection to Portfolio (Section 4):** The detected market regime (Bull/Bear/Sideways) should inform strategy selection. During a detected Bear regime, the Min Variance or Min CVaR strategies may be more appropriate than Max Sharpe. This module provides the regime context for strategic asset allocation shifts.
- **Regime Detection to Monte Carlo (Section 6):** The GMM-based regime detection here complements the Monte Carlo module's own regime framework. Transition probabilities between regimes estimated in this module can calibrate the Monte Carlo model's regime-switching dynamics.
- **Anomaly Detection to Risk Dashboard (Section 9):** Detected anomalous trading days can be cross-referenced with macro risk events. If anomalies cluster around high-risk periods, they likely reflect systemic stress; if they occur during low-risk periods, they may indicate asset-specific events.
- **PCA to Portfolio (Section 4):** PCA factor loadings reveal the effective dimensionality of the portfolio. If PC1 explains 80%+ of variance, the portfolio has concentrated factor exposure regardless of how many assets it holds. This insight can motivate portfolio rebalancing toward greater factor diversification.
- **Clustering to Bubble Detection (Section 5):** Assets clustered together should exhibit similar bubble characteristics. If one asset in a cluster shows elevated bubble scores while others do not, it may be experiencing idiosyncratic speculation rather than a sector-wide bubble.

11.1 Asset Clustering

Assets are clustered based on their statistical properties to identify natural groupings and peer relationships.

Feature Vector per Asset

Each asset is represented by a 6-dimensional feature vector: annualized return, annualized volatility, skewness, kurtosis, Sharpe ratio, and maximum drawdown.

K-Means Clustering

$$\min \sum ||x_i - \mu_k||^2 \text{ over } k \text{ clusters}$$

K-Means partitions assets into k hard clusters by minimizing within-cluster sum of squares. The optimal k is determined by silhouette analysis or user specification.

Gaussian Mixture Model (GMM)

GMM provides probabilistic (soft) clustering where each asset has a probability of belonging to each cluster. This is more flexible than K-Means when cluster shapes are elliptical rather than spherical.

PCA Visualization

Principal Component Analysis reduces the 6-dimensional feature space to 2D for visualization. Cluster assignments are color-coded on the PCA scatter plot, allowing visual assessment of cluster separation quality.

11.2 Market Regime Detection

QuantLab uses Gaussian Mixture Models in a Hidden Markov-style framework to detect market regimes from log-return data.

- Fit GMM with `n_components=2` (Bull/Bear) or 3 (Bull/Bear/Sideways) on rolling log-returns.
- Classify the current regime based on the latest observation's cluster assignment.
- Compute regime-specific statistics: mean return, volatility, and historical duration for each regime.
- Regime transitions are tracked to estimate transition probabilities between states.

11.3 Anomaly Detection

Isolation Forest identifies anomalous trading days based on return and volume patterns.

$$\text{contamination} = 0.05 \text{ (expected 5\% anomaly rate)}$$

The algorithm isolates observations by randomly selecting features and split values. Anomalies (outliers) are isolated with fewer splits and receive lower anomaly scores. Detected anomalies are visually overlaid on price charts as highlighted markers.

11.4 PCA Factor Analysis

Principal Component Analysis on the return matrix extracts latent factors driving portfolio returns:

- **Factor Loadings:** Show which assets drive each principal component. PC1 typically represents broad market exposure; PC2 captures sector rotation or risk-on/risk-off dynamics.
- **Variance Explained:** Cumulative variance plot determines the effective dimensionality of the return space — how many independent risk factors drive the portfolio.
- **Dimensionality Reduction:** PCA factors can serve as inputs to regression models, reducing multicollinearity among correlated assets.

12. Sentiment Analysis

QuantLab's sentiment analysis module provides keyword-based news sentiment scoring using a curated financial lexicon. It processes recent headlines from Yahoo Finance's news API to generate directional market sentiment indicators.

Purpose & Role in the Investment Workflow

Sentiment analysis introduces an alternative data dimension that captures information not embedded in price or volume data. News headlines reflect market narrative, institutional commentary, and event-driven catalysts that can precede or explain price movements. For institutional investors, sentiment serves as a contrarian indicator at extremes (overwhelming bullishness often precedes corrections) and as a confirmation signal during trends (sustained positive sentiment supports momentum). The keyword-based approach is deliberately chosen over deep learning NLP for three reasons: it is fully interpretable (you can see exactly which words drive the score), it requires no GPU or model training, and it executes in real time. For hedge funds with proprietary NLP models, this module serves as a fast baseline that can be compared against more sophisticated sentiment pipelines.

Cross-Module Interactions

- **Sentiment to Bubble Detection (Section 5):** Extreme bullish sentiment during a detected bubble phase creates the classic 'euphoria' signal that historically precedes sharp corrections. Conversely, bearish sentiment during a low-bubble-score period may signal a buying opportunity.
- **Sentiment to Technical Analysis (Section 7):** Divergence between sentiment and technicals is a powerful signal. If sentiment is strongly bullish but RSI shows overbought and MACD is crossing bearishly, the technical signals may be leading the narrative shift.
- **Sentiment to ML Predictions (Section 10):** Sentiment scores could serve as additional features in the ML prediction pipeline. While not currently included in the default feature set, the aggregated sentiment score is available for custom feature engineering.
- **Sentiment to Risk Dashboard (Section 9):** News-driven fear events (geopolitical crises, central bank surprises) typically show up in sentiment before they fully propagate into VIX and other risk indicators. Sentiment provides an early-warning lead time.
- **Sentiment to Export (Section 13):** Article-level sentiment scores, distribution charts, and top keyword frequencies are included in all three export formats, providing a qualitative narrative layer alongside quantitative analytics.

12.1 News Fetching

Recent news articles are retrieved via the `yf.Ticker(symbol).news` API, which returns up to 20 articles per ticker. Article metadata includes title, publisher, link, and publication timestamp.

12.2 Keyword-Based Sentiment Scoring

QuantLab uses a lexicon-based approach with 29 positive and 28 negative financial keywords to score article sentiment.

Sentiment Score Formula

$$S = (\text{count}_{\text{positive}} - \text{count}_{\text{negative}}) / (\text{count}_{\text{positive}} + \text{count}_{\text{negative}})$$

Score Range	Classification	Interpretation
S > 0.1	Bullish	Positive market sentiment
-0.1 <= S <= 0.1	Neutral	Mixed or balanced signals
S < -0.1	Bearish	Negative market sentiment

Sample Lexicon Keywords

Positive Words	Negative Words
growth, surge, rally, upgrade, beat	decline, crash, downgrade, miss, risk
profit, bullish, outperform, strong	loss, bearish, underperform, weak
innovation, dividend, breakout, record	recession, default, inflation, layoff

12.3 Aggregation

Individual article scores are aggregated through several methods:

- **Overall Sentiment:** Arithmetic mean of all article scores.
- **Distribution Chart:** Count of Bullish, Neutral, and Bearish articles displayed as a bar chart.
- **Source-Level Aggregation:** Average sentiment grouped by publisher, revealing potential source bias.
- **Top Keyword Frequency:** Term frequency (TF) analysis of the most common words across all articles to identify dominant narratives.

Institutional Relevance: While lexicon-based sentiment is a simplified approach compared to transformer-based NLP models, it provides fast, interpretable, and reproducible sentiment signals suitable for real-time monitoring and alert systems.

13. Export System

QuantLab provides three export formats for distributing analysis results: PDF reports, presentation slide decks, and Excel workbooks. All exports are generated client-side using Python libraries and delivered via Streamlit download buttons.

Purpose & Role in the Investment Workflow

The Export System transforms QuantLab from an interactive analysis tool into a production-grade reporting platform. In institutional finance, analysis is only as valuable as its ability to be communicated to decision-makers. The PDF report serves investment committees and compliance teams who require a printable, audit-ready record of analysis. The slide deck serves portfolio managers presenting to clients or internal stakeholders. The Excel workbook serves quantitative analysts who need to perform custom calculations on top of QuantLab's outputs. Each format captures the complete analytical pipeline — from raw performance metrics through valuation, optimization, bubble detection, technicals, macro risk, ML predictions, options analysis, clustering, and sentiment — ensuring that the exported artifact is a self-contained investment research document.

Cross-Module Interactions

- **Export from All Modules (Sections 2-12):** The export system is the terminal consumer of every other module. It reads computed metrics, optimized weights, bubble scores, technical indicators, risk scores, ML predictions, options Greeks, cluster assignments, and sentiment scores from Streamlit session state.
- **Export with Configuration (Section 14):** The 'Export Sections' setting allows users to selectively include or exclude specific analysis sections from exports, enabling customized reports for different audiences (e.g., a risk-focused report that omits sentiment).
- **PDF/Slides use Matplotlib:** While the interactive app uses Plotly for charts, exports render static Matplotlib versions because fpdf2 requires raster images. This dual-rendering approach means chart styling must be maintained in both Plotly and Matplotlib codepaths.
- **Excel with NaN Handling:** The `xlswriter` engine uses `nan_inf_to_errors=True` to handle NaN and infinity values that commonly appear in financial calculations (e.g., Sharpe ratio when volatility is zero, or Kelly weights when the covariance matrix is singular).

13.1 PDF Report (fpdf2)

The PDF export generates a multi-page A4 portrait document covering all analysis sections in approximately 15 pages.

- Custom header/footer with QuantLab branding and page numbers
- Matplotlib charts rendered as embedded PNG images
- Formatted data tables for all metric sections
- Section-by-section analysis including metrics, valuations, portfolio allocations, bubble scores, technicals, macro indicators, ML predictions, options analysis, and sentiment scores

13.2 Slide Deck (fpdf2)

The slide deck export creates a landscape 16:9 presentation (297x167mm) with 16 slides:

Slide #	Content	Description
1	Cover	Title, date, tickers analyzed
2	Portfolio Overview	Asset list, date range, data summary
3	Performance Metrics	Returns, volatility, Sharpe ratios
4-5	Valuation Models	CAPM, WACC, DCF, Fama-French, APT results
6-7	Portfolio Optimization	Weights, efficient frontier, strategy comparison
8	Bubble Detection	Composite scores, MMV ratios, regime classification
9	Technical Analysis	SMA, MACD, RSI, Bollinger Bands
10	Macro & Risk	Risk score, VIX, yield curve, correlations
11	ML Predictions	Model performance, feature importance
12	Options Analysis	Pricing, Greeks, strategy payoffs
13	Risk Dashboard	Composite risk score, indicator panel
14	Clustering	Asset clusters, regime detection, PCA
15	Sentiment	News sentiment scores, distribution
16	Key Takeaways	Summary findings, recommendations

13.3 Excel Workbook (xlsxwriter)

The Excel export creates a multi-sheet workbook with formatted data tables, enabling further analysis in spreadsheet applications.

Sheet	Contents
Prices	Historical adjusted close prices for all tickers
Metrics	Performance metrics: returns, volatility, Sharpe, drawdown
Valuation	CAPM, WACC, DCF, Fama-French, APT model outputs
Portfolio	Optimization weights for all 8 strategies
Bubble	Composite scores, MMV ratios, long memory d
Technicals	SMA, EMA, MACD, RSI, Bollinger Bands time series
Macro	VIX, DXY, Gold, Oil, Treasury yields

Sheet	Contents
ML	Predictions, actuals, model accuracy metrics
Options	Pricing, Greeks, strategy payoffs
Risk	Composite risk score components
Clustering	Cluster assignments, regime labels, PCA loadings
Sentiment	Article scores, aggregated sentiment, keyword frequencies

The workbook uses `nan_inf_to_errors=True` option to handle NaN and infinity values gracefully. Formatted headers, number formats, and conditional formatting are applied for professional presentation.

14. Configuration & Advanced Settings

QuantLab exposes comprehensive configuration options through the Streamlit sidebar, organized into functional groups:

Purpose & Role in the Investment Workflow

The Configuration system is what makes QuantLab adaptable to different investment styles, risk tolerances, and analytical workflows. A risk-averse pension fund manager will configure very differently from an aggressive crypto hedge fund: different Monte Carlo simulation counts, different ML training windows, different clustering sensitivity, and different export sections. The Quick Presets feature (22 options spanning Tech, Crypto, Forex, Commodities, and more) provides instant workflow templates, while the Advanced Settings sections give granular control over every analytical parameter. Critically, all settings propagate to their respective modules in real time — changing the Monte Carlo confidence level from 95% to 99% immediately recomputes simulation confidence bands without requiring a full page reload.

Cross-Module Interactions

- **Configuration to Monte Carlo (Section 6):** Simulation count, forecast horizon, confidence level, and method selection (GBM vs. Behavioral Agent) directly control the Monte Carlo engine's behavior.
- **Configuration to Portfolio (Section 4):** Bubble-aware toggle, penalty factor, benchmark selection, risk-free rate override, and rebalancing frequency parametrize the portfolio optimization pipeline.
- **Configuration to ML (Sections 10-11):** Training period length, clustering method (K-Means vs. GMM), anomaly sensitivity, and cluster count govern the machine learning modules.
- **Configuration to Export (Section 13):** Display settings (chart height, table row limit) affect how visualizations render in exports. The 'Export Sections' toggle controls which analytical sections appear in the generated reports.
- **Quick Presets to Market Data (Section 2):** Each preset automatically populates the ticker selection with curated asset lists (e.g., 'Forex Majors' loads EURUSD=X, GBPUSD=X, etc.), triggering the entire data pipeline to refresh for the new asset universe.

14.1 Monte Carlo Settings

Setting	Default	Range	Description
Simulations	1000	100-10000	Number of Monte Carlo paths
Forecast Days	252	21-756	Simulation horizon (trading days)
Confidence Level	95%	80-99%	Confidence interval width
Method	Behavioral Agent	GBM / Agent	Simulation engine selection

14.2 Portfolio Settings

Setting	Default	Description
Bubble-Aware	Enabled	Apply bubble penalty to optimization
Penalty Factor	0.5	Bubble score weight in objective function
Benchmark	SPY	Benchmark ticker for relative performance
Risk-Free Override	Auto (^IRX)	Manual risk-free rate override
Rebalancing	Monthly	Portfolio rebalancing frequency

14.3 Machine Learning Settings

Setting	Default	Description
Training Period	3 years	Historical data window for model training
Clustering Method	K-Means	K-Means or Gaussian Mixture Model
Anomaly Sensitivity	0.05	Isolation Forest contamination parameter
Number of Clusters	3	Target cluster count

14.4 Display Settings

Setting	Default	Description
Chart Height	500px	Plotly chart height in pixels
Table Row Limit	50	Maximum rows displayed in data tables
Export Sections	All	Selective section inclusion for exports

Appendix A: Supported Asset Classes & Tickers

A.1 Stocks & ETFs

Standard ticker symbols as traded on major US exchanges. Examples:

Category	Tickers
Mega-Cap Tech	AAPL, MSFT, GOOGL, AMZN, NVDA, META, TSLA
Financials	JPM, BAC, GS, MS, BRK-B, V, MA
Healthcare	JNJ, UNH, PFE, ABBV, MRK, TMO, LLY
ETFs (Broad)	SPY, QQQ, IWM, VTI, VOO, DIA
ETFs (Sector)	XLF, XLK, XLE, XLV, XLI, XLRE
ETFs (Fixed Income)	TLT, IEF, SHY, AGG, BND, HYG, LQD

A.2 Forex

Currency pairs use the =X suffix:

Pair	Ticker	Description
EUR/USD	EURUSD=X	Euro vs US Dollar
GBP/USD	GBPUSD=X	British Pound vs US Dollar
USD/JPY	JPY=X	US Dollar vs Japanese Yen
USD/CHF	CHF=X	US Dollar vs Swiss Franc
AUD/USD	AUDUSD=X	Australian Dollar vs US Dollar

A.3 Commodities

Commodity futures use the =F suffix:

Commodity	Ticker	Exchange
Gold	GC=F	COMEX
Crude Oil (WTI)	CL=F	NYMEX
Silver	SI=F	COMEX

Commodity	Ticker	Exchange
Natural Gas	NG=F	NYMEX
Copper	HG=F	COMEX
Corn	ZC=F	CBOT

A.4 Cryptocurrency

Crypto assets use the -USD suffix:

Asset	Ticker	Market Cap Tier
Bitcoin	BTC-USD	Large Cap
Ethereum	ETH-USD	Large Cap
Solana	SOL-USD	Mid Cap
Cardano	ADA-USD	Mid Cap
Dogecoin	DOGE-USD	Mid Cap
Polygon	MATIC-USD	Mid Cap

Appendix B: Dependencies & Requirements

B.1 Core Dependencies

Package	Version	Role
streamlit	>= 1.28	Web application framework
yfinance	>= 0.2.31	Yahoo Finance data API
pandas	>= 2.0	DataFrames & time series
numpy	>= 1.24	Numerical computation
scipy	>= 1.11	Optimization & statistical tests
scikit-learn	>= 1.3	Machine learning models
plotly	>= 5.18	Interactive visualizations
matplotlib	>= 3.8	Static chart generation for exports
ta	>= 0.11	Technical analysis indicators
statsmodels	>= 0.14	Ljung-Box test, time series analysis
fpdf2	>= 2.7	PDF report & slide generation
xlsxwriter	>= 3.1	Excel workbook generation

B.2 Installation

```
# Create virtual environment
python -m venv quantlab-env
source quantlab-env/bin/activate

# Install dependencies
pip install -r requirements.txt

# Run application
streamlit run quantlab.py
```

B.3 System Requirements

Requirement	Minimum	Recommended
Python	3.9	3.11+
RAM	4 GB	8 GB+

Requirement	Minimum	Recommended
Internet	Required	Broadband for real-time data
Browser	Chrome/Firefox	Chrome (latest)

End of Document — QuantLab Technical Documentation

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